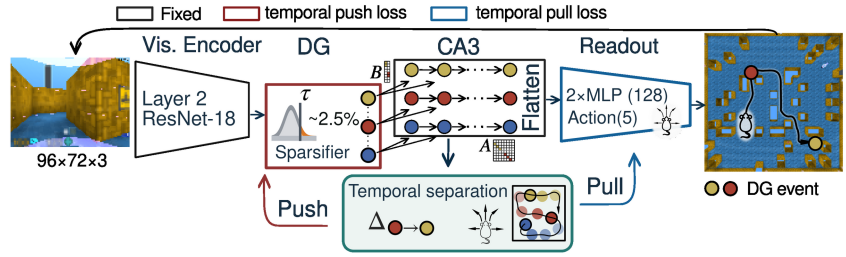
A Push–pull navigation architecture

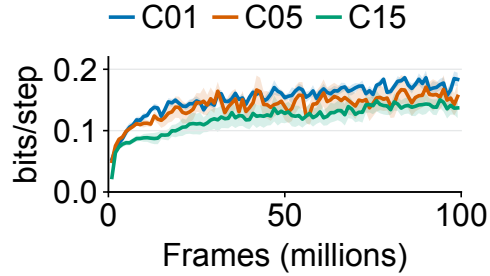


Fixed: ResNet-18 visual trunk and CA3 memory.

Learned: DG projection, batch normalization, readout.

Red push trains DG; blue pull shapes the policy.

B Spatial coding emerges



C Example DG place fields

